

- Description

This module manages a smart water pump system, ensuring that:

1. **Water pressure in the pipeline is continuously monitored and adjusted to maintain stable supply.**
2. **If the pressure exceeds the safety threshold, the pump reduces power to prevent pipe damage.**
3. **If the pressure drops abnormally, a leak warning is triggered for inspection.**
4. **A log of all pressure readings, pump adjustments, and leak alerts is maintained for operational optimization.**

This system helps prevent pipeline failures, optimize water distribution, and enhance maintenance efficiency.

- Specification

module SmartWaterPumpManagement

var currentPressure, maxPressure, minPressure: real

var pumpActive: boolean

var log: sequence of string

process updatePressure(changeAmount: real)

functional scenario form:

currentPressure $\geq$ minPressure && log = log $\oplus$ ("Pressure Updated: " + currentPressure + " bar, Status Normal."), Water pressure updated successfully.

||

currentPressure > maxPressure && pumpActive = false, log = log $\oplus$ ("High Pressure Warning: " + currentPressure + " bar, Reducing Pump Power."), High pressure warning triggered, pump power reduced.

||

currentPressure < minPressure && pumpActive = true, log = log $\oplus$ ("Leak Alert: " + currentPressure + " bar, Inspection Required."), Leak detected, inspection required.

end process

end module

- Java Code

```
package code;
import java.util.*;
class SmartWaterPump {
    private double maxPressure;
    private double minPressure;
    private double currentPressure;
    private boolean pumpActive;
    private List<String> log;
    public SmartWaterPump(double maxPressure, double minPressure, double initialPressure) {
        this.maxPressure = maxPressure;
```

```

        this.minPressure = minPressure;
        this.currentPressure = initialPressure;
        this.pumpActive = true;
        this.log = new ArrayList<>();
    }
    public void updatePressure(double changeAmount) {
        currentPressure += changeAmount;
        managePumpOperation();
        String message = String.format("Current Pressure: %.2f bar", cu
rrentPressure);
        if (currentPressure > maxPressure) {
            message += " | WARNING: High Pressure! Reducing Pump Power.
";
        } else if (currentPressure < minPressure) {
            message += " | ALERT: Possible Leak Detected! Inspection Re
quired.";
        } else {
            message += " | Pressure is Normal.";
        }
        System.out.println(message);
        log.add(message);
    }
    private void managePumpOperation() {
        if (currentPressure > maxPressure) {
            pumpActive = false;
        } else if (currentPressure < minPressure) {
            pumpActive = true;
        }
    }
    public void printLog() {
        System.out.println("\nWater Pump Pressure Log:");
        for (String logEntry : log) {
            System.out.println(logEntry);
        }
    }
}

public class c52_SmartWaterPumpSystem {
    public static void main(String[] args) {
        SmartWaterPump pump = new SmartWaterPump(8.0, 3.0, 5.0);
        // testcase-VT:
        //     pump.updatePressure(2.5);
        //     pump.updatePressure(-3.0);
        //     pump.updatePressure(-2.0);
        //     pump.updatePressure(4.0);
    }
}

```

```
//      pump.updatePressure(3.5);
//      pump.printLog();

//      // testcase-FT:
//      pump.updatePressure(-1.7976931348623157E308);
//      pump.updatePressure(0.0);
//      pump.updatePressure(1.7976931348623157E308);
//      pump.updatePressure(8.530048910211935E300);
//      pump.updatePressure(4.665666536097863E293);
//      pump.printLog();

//      // testcase-Z3:
//      pump.updatePressure(2);
//      pump.updatePressure(-3);
//      pump.updatePressure(-2);
//      pump.updatePressure(4);
//      pump.updatePressure(3);
//      pump.printLog();

//      // testcase-UVT:
//      pump.updatePressure(2.5);
//      pump.updatePressure(-3.0);
//      pump.updatePressure(-2.0);
//      pump.updatePressure(4.0);
//      pump.updatePressure(3.5);
//      pump.printLog();
    }
}
```

- Coverage report

1. Test case number: 5
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.2 %	10	832	842
▼ src	1.2 %	10	832	842
▼ code	1.2 %	10	832	842
▼ c52_SmartWaterPumpSystem.java	100.0 %	10	0	10
> c52_SmartWaterPumpSystem		0	0	0
> SmartWaterPump	100.0 %	10	0	10

3. Vibration testing instruction coverage: 97.8%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.7 %	135	20,052	20,187
▼ src	0.7 %	135	20,052	20,187
▼ code	0.7 %	135	20,052	20,187
▼ c52_SmartWaterPumpSystem.java	97.8 %	135	3	138
> c52_SmartWaterPumpSystem	89.3 %	25	3	28
> SmartWaterPump	100.0 %	110	0	110

4. Fuzz testing branch coverage: 80%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	1.0 %	8	834	842
✓ src	1.0 %	8	834	842
✓ code	1.0 %	8	834	842
✓ c52_SmartWaterPumpSystem.java	80.0 %	8	2	10
> c52_SmartWaterPumpSystem		0	0	0
> SmartWaterPump	80.0 %	8	2	10

5. Fuzz testing instruction coverage: 95.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.7 %	132	20,055	20,187
✓ src	0.7 %	132	20,055	20,187
✓ code	0.7 %	132	20,055	20,187
✓ c52_SmartWaterPumpSystem.java	95.7 %	132	6	138
> c52_SmartWaterPumpSystem	89.3 %	25	3	28
> SmartWaterPump	97.3 %	107	3	110

6. Z3 branch coverage:100 %

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	1.2 %	10	832	842
✓ src	1.2 %	10	832	842
✓ code	1.2 %	10	832	842
✓ c52_SmartWaterPumpSystem.java	100.0 %	10	0	10
> c52_SmartWaterPumpSystem		0	0	0
> SmartWaterPump	100.0 %	10	0	10

7. Z3 instruction coverage: 97.8%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.7 %	135	20,052	20,187
✓ src	0.7 %	135	20,052	20,187
✓ code	0.7 %	135	20,052	20,187
✓ c52_SmartWaterPumpSystem.java	97.8 %	135	3	138
> c52_SmartWaterPumpSystem	89.3 %	25	3	28
> SmartWaterPump	100.0 %	110	0	110

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	1.2 %	10	832	842
✓ src	1.2 %	10	832	842
✓ code	1.2 %	10	832	842
✓ c52_SmartWaterPumpSystem.java	100.0 %	10	0	10
> c52_SmartWaterPumpSystem		0	0	0
> SmartWaterPump	100.0 %	10	0	10

9. Unimproved V-method instruction coverage: 97.8%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.7 %	135	20,052	20,187
✓ src	0.7 %	135	20,052	20,187
✓ code	0.7 %	135	20,052	20,187
✓ c52_SmartWaterPumpSystem.java	97.8 %	135	3	138
> c52_SmartWaterPumpSystem	89.3 %	25	3	28
> SmartWaterPump	100.0 %	110	0	110